Week 5 · Lab 2

Version A

Graded Practice:

`?:`, `switch` and `while`

10 problems from LeetCode · 10 points each · checked by the TA at your desk

Week: 5 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lecture 4 — the ternary operator `?:`, `switch`/`case`, the `while` loop, `break`/`continue`

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. There are **10 problems, 10 points each** (100 in total): problems 1–3 are **easy**, 4–7 **medium** and 8–10 **hard**. Solve them in any order; a problem with no attempt earns 0 points. There is enough work for the whole session — keep going until the end.
4. Under each title, **Practises:** says which part of Lecture 4 the problem trains.
5. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard (a list of numbers or a word is read one item at a time, because arrays and strings come later). On LeetCode itself the answer is written as a *function* inside `class Solution`; functions come next week.

Quick reference

Ternary operator `condition ? a : b` is a *value*. Put it in parentheses inside `cout`.

```
int mx = a > b ? a : b;
cout << (n % 2 == 0 ? "even" : "odd");
d = d < 0 ? -d : d;                  // |d|
```

`switch` for `int`/`char` values compared with constants. `break` leaves; without it execution falls through into the next `case`. Several labels may share one body.

```
switch (c) {
    case 'a': case 'e': case 'i':
    case 'o': case 'u': vowels++; break;
    case ' ':           spaces++; break;
    default:            others++;
}
```

`while`: initialise, test, update The test runs before every pass, so the body may run zero times.

```
int i = 1;                       // initialise
while (i <= n) {                 // test
    total += i;
    i++;                         // update
}
```

Reading *n* items without an array read one, use it, forget it:

```
int n;  cin >> n;
int best = -1000000000;
int i = 0;
while (i < n) {
    int x;  cin >> x;            // also: char c; cin >> c;
    if (x > best) { best = x; }  // running maximum
    i++;
}
```

A word given as `5 hello` is read the same way: `n`, then `n` single `char`s.

Digit by digit

```
while (n > 0) {
    int d = n % 10;              // last digit
    n /= 10;                     // drop it
}
```

`break` and `continue` `break` leaves the innermost loop (or `switch`) at once; `continue` jumps back to the test. In a `while`, do the update *before* a `continue`, or the loop never ends. A `while` inside a `while`: reset the inner loop's variables inside the outer loop.

Classic bugs a missing `break`; no update, or `while (...);` — an endless loop; `i != 10` with `i += 3`; `<` versus `<=`; a product that starts at 0; `int` overflow (use `long long`).

EASY

Problems 1–3 · 30 points

Short programs: one idea each. Test every branch, not just the examples.

1 Sign of the Product of an Array 10 pts

LeetCode 1822 · `leetcode.com/problems/sign-of-the-product-of-an-array`

Practises: a `while` loop over the input with `continue`, and `?:`.

Read n ($1 \le n \le 1000$), then n integers, each from -100 to 100. Print `1` if their product is positive, `-1` if it is negative and `0` if it is zero.

Input	Output	Why
`6 -1 -2 -3 -4 3 2`	`1`	The product is 144.
`5 1 5 0 2 -3`	`0`	
`5 -1 1 -1 1 -1`	`-1`	

Hint: do not *multiply the numbers — 1000 numbers of size 100 overflow any type. A positive number changes nothing — skip it with* `continue` *(update the counter* before *it). Keep only the sign: a zero makes it 0, a negative number flips it.*

TA: ____ / 10 pts initials _____

2 Number of Steps to Reduce a Number to Zero 10 pts

LeetCode 1342 · `leetcode.com/problems/number-of-steps-to-reduce-a-number-to-zero`

Practises: a `while` loop, and `?:` for the step.

Read `num` ($0 \le$ `num` $\le 10^6$). In one step, an even number is divided by 2 and an odd number has 1 subtracted. Print how many steps it takes to reach 0.

Input	Output	Why
`14`	`6`	$14 \to 7 \to 6 \to 3 \to 2 \to 1 \to 0$.
`8`	`4`	$8 \to 4 \to 2 \to 1 \to 0$.
`123`	`12`	

TA: ____ / 10 pts initials _____

3 Subtract the Product and Sum of Digits of an Integer 10 pts

LeetCode 1281 · `leetcode.com/problems/subtract-the-product-and-sum-of-digits-of-an-integer`

Practises: a `while` loop over the digits (`% 10`, `/ 10`).

Read n ($1 \le n \le 10^5$). Print (product of its digits) − (sum of its digits).

Input	Output	Why
`234`	`15`	$2 \cdot 3 \cdot 4 - (2 + 3 + 4) = 24 - 9$.
`4421`	`21`	$32 - 11$.

TA: ____ / 10 pts initials _____

MEDIUM

Problems 4–7 · 40 points

A few cases to get right. Test the boundaries.

4 Alternating Digit Sum 10 pts

LeetCode 2544 · `leetcode.com/problems/alternating-digit-sum`

Practises: a `while` loop over the digits, with a sign that flips.

Read a positive integer n ($1 \le n \le 10^9$). Give its *leftmost* digit a plus sign, the next a minus, and so on, alternating. Print the sum.

Input	Output	Why
`521`	`4`	+5 − 2 + 1.
`111`	`1`	+1 − 1 + 1.
`886996`	`0`	+8 − 8 + 6 − 9 + 9 − 6.

Hint: the digit loop meets the digits from the right. *Add them with signs +, −, +, . . . from the right and count them. If the number of digits is even, the leftmost digit got a minus — so flip the whole sum.*

TA: ____ / 10 pts initials ____

5 Roman to Integer 10 pts

LeetCode 13 · `leetcode.com/problems/roman-to-integer`

Practises: `switch` inside a loop over characters.

Read n ($1 \le n \le 15$), then a valid Roman numeral of n symbols (written without spaces) between 1 and 3999, and print its value. `I` = 1, `V` = 5, `X` = 10, `L` = 50, `C` = 100, `D` = 500, `M` = 1000. A symbol written *before* a larger one is subtracted: `IV` = 4, `XC` = 90, `CM` = 900.

Input	Output	Why
`3 III`	`3`	
`5 LVIII`	`58`	50 + 5 + 3.
`7 MCMXCIV`	`1994`	1000 + 900 + 90 + 4.

Hint: read the symbols one `char` *at a time and turn each into its value with a* `switch`*. When the current value is bigger than the previous one, the previous one should have been subtracted, not added — correct for that.*

TA: ____ / 10 pts initials ____

6 Happy Number 10 pts

LeetCode 202 · `leetcode.com/problems/happy-number`

Practises: `while (true)` with `break`, and a `while` loop inside it.

Start with a positive integer n ($1 \le n \le 2^{31} - 1$) and replace it again and again by the sum of the squares of its digits. n is *happy* if this reaches 1. Print `true` if n is happy, otherwise `false`.

Input	Output	Why
`19`	`true`	1 + 81 = 82, 64 + 4 = 68, 36 + 64 = 100, 1 + 0 + 0 = 1.
`2`	`false`	2 → 4 → 16 → 37 → 58 → 89 → 145 → 42 → 20 → 4 → . . . for ever.

Hint: a number that is not happy never reaches 1 — it falls into the cycle in the second example, which always passes through 4. So loop with `while (true)` *and* `break` *as soon as you reach 1 or 4. The inner* `while` *computes the sum of squared digits.*

TA: ____ / 10 pts initials ____

7 Best Time to Buy and Sell Stock 10 pts

LeetCode 121 · `leetcode.com/problems/best-time-to-buy-and-sell-stock`

Practises: one pass with a running minimum.

Read n ($1 \le n \le 10^5$), then the price of a share on each of n days (each from 0 to 10^4). You may buy on one day and sell on a *later* day. Print the largest possible profit, or 0 if no trade makes money.

Input	Output	Why
`6 7 1 5 3 6 4`	`5`	Buy at 1 (day 2), sell at 6 (day 5).
`5 7 6 4 3 1`	`0`	Prices only fall.

Hint: you cannot store the prices (no arrays yet), and you do not need to. If you sell today, the best day to have bought is the cheapest day so far.

TA: _____ / 10 pts initials _____

HARD

Problems 8–10 · 30 points

More cases and more care. Read the hint before you start.

8 Sequential Digits 10 pts

LeetCode 1291 · `leetcode.com/problems/sequential-digits`

Practises: `while` loops inside `while` loops.

A number has *sequential digits* if each digit is one more than the digit before it, e.g. `345` or `6789`. Read `low` and `high` ($10 \le$ `low` $\le$ `high` $\le 10^9$) and print, in increasing order and separated by spaces, all numbers with sequential digits in the range. Print `none` if there are none.

Input	Output	Why
`100 300`	`123 234`	
`1000 6000`	`1234 2345 3456 4567 5678`	

Hint: do not test every number up to 10^9 — build them. Choose the length (2 to 9) and the first digit, then build the number digit by digit. Shorter numbers are smaller, so loop over the length first.

TA: ____ / 10 pts initials _____

9 Reverse Integer 10 pts

LeetCode 7 · `leetcode.com/problems/reverse-integer`

Practises: a `while` loop, `long long`, and a range check.

Read an `int` x (negative values included). Print x with its digits reversed, keeping the sign. If the reversed number does not fit in an `int` (-2^{31} to $2^{31} - 1$), print `0`.

Input	Output	Why
`123`	`321`	
`-123`	`-321`	
`120`	`21`	

Hint: build the reversed number in a `long long`*, then check its range. In C++,* `-123 % 10` *is* `-3`*, so negative numbers need no special case.*

TA: ____ / 10 pts initials _____

10 Maximum Subarray 10 pts

LeetCode 53 · `leetcode.com/problems/maximum-subarray`

Practises: one pass with `?:` — a famous idea (Kadane).

Read n ($1 \le n \le 10^5$), then n integers (each from -10^4 to 10^4). Print the largest sum of a block of *consecutive* numbers (at least one number).

Input	Output	Why
`9 -2 1 -3 4 -1 2 1 -5 4`	`6`	$4 + (-1) + 2 + 1$.
`1 1`	`1`	
`5 5 4 -1 7 8`	`23`	All five.

Hint: keep `current`*, the best sum of a block that* ends at the number just read. *Either the block continues (*`current + x`*) or a new block starts at* `x` *— when is starting again better?*

TA: ____ / 10 pts initials _____

Score sheet — Version A

	Problem	Level	Points	Earned	TA
1	Sign of the Product of an Array (LC 1822)	Easy	10		
2	Number of Steps to Reduce a Number to Zero (LC 1342)	Easy	10		
3	Subtract the Product and Sum of Digits of an Integer (LC 1281)	Easy	10		
4	Alternating Digit Sum (LC 2544)	Medium	10		
5	Roman to Integer (LC 13)	Medium	10		
6	Happy Number (LC 202)	Medium	10		
7	Best Time to Buy and Sell Stock (LC 121)	Medium	10		
8	Sequential Digits (LC 1291)	Hard	10		
9	Reverse Integer (LC 7)	Hard	10		
10	Maximum Subarray (LC 53)	Hard	10		
	Total		**100**		

After the lab

Every problem here is on LeetCode under the number shown. Here the input is read one item at a time; on LeetCode it arrives as an array or a string. Once functions are covered, rewrite your solutions as a LeetCode `class Solution` and submit them.

End of Week 5 · Lab 2 · Version A